



Research paper

Advancing fault diagnosis in industrial systems: The power of V-nets for managing complex event sequences

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ABSTRACT

This paper introduces V-nets, an innovative formal model developed to enhance fault diagnosis and management of complex Discrete Event Systems (DES), particularly in industrial applications. The V-net model addresses key challenges in traditional models such as Petri nets and automata, including the management of simultaneous events, minimizing false positives, and improving the recognition of partial event sequences. Methods: This study builds upon a comparative analysis of existing fault diagnosis techniques, integrating V-nets with automata to form a robust framework for supervising and diagnosing industrial systems. Through a case study of the Reactor Startup Sequence of a Pressurized Water Reactor (PWR), the research highlights the practical utility of V-nets in real-time fault detection and system monitoring. Findings: The experimental results demonstrate that V-nets significantly reduce false positives and enhance system flexibility when dealing with complex event sequences. By evaluating over 100,000 event sequences using Monte Carlo methods, the study confirms the accuracy and scalability of the V-net model in industrial settings. Novelty: This research pioneers the development of V-nets as a superior alternative to traditional models for fault diagnosis in DES, offering a framework capable of handling simultaneous events and complex time constraints. The proposed model's potential for real-world implementation opens new avenues for advancing industrial process supervision. The flexibility and scalability of V-nets represent a critical advancement in the formal methods for industrial systems, positioning this model as a promising tool for future industrial applications and automatic V-net generation in real-time scenarios.

1. Introduction

The advancement of methodologies for fault detection and diagnosis in discrete event systems (DES) can benefit from recent innovations in machine learning and deep learning models. Several studies have highlighted the potential of these computational frameworks in improving event sequence detection and anomaly recognition, both of which are essential in DES supervision. For instance, [Elmessery et al. \(2024\)](#) presents a semantic segmentation approach using SegFormer for microbial alterations, showcasing the capability of segmentation models to detect complex patterns in large datasets. Their work emphasizes the importance of leveraging advanced deep learning techniques for pattern recognition, which is highly relevant for identifying abnormal sequences in DES. Similarly, [Mostafa et al. \(2024\)](#) focuses on feature reduction techniques for predicting hepatocellular carcinoma using machine learning algorithms. The feature reduction approach they explore not only simplifies the dataset but also enhances classification accuracy. Such methodologies could inspire the development of streamlined approaches in DES event sequence analysis, helping to reduce the complexity of the system while maintaining diagnostic precision.

Furthermore, [El Koshiry et al. \(2024\)](#) applies deep learning techniques, particularly a pre-trained GloVe model combined with focal loss, to detect cyberbullying. Their research demonstrates how specific model architectures can be tailored to handle imbalanced data, a challenge that is also common in fault detection in DES where some event sequences occur more frequently than others. [Hassan et al. \(2024\)](#) introduces a model for optimizing disease classification through language model analysis of symptoms. Their model optimization approach could provide insights into how language models might be adapted for DES event sequence analysis, enhancing fault detection by improving the recognition of subtle anomalies. Additionally, [Abulohoom and Elnagar \(2024\)](#) presents a study on toxicity detection in Arabic text using deep learning techniques, further demonstrating the versatility of language models for detecting anomalies in textual datasets, which could be relevant for error detection in DES event sequences. [Shams et al. \(2024\)](#) focuses on acoustic data detection in large-scale datasets, introducing a Self-Attention Layer within a Convolutional Neural Network (SACNN) for detecting emergency vehicle sirens and road noise. The emphasis

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on precision and computational efficiency in this study could inform future DES fault detection systems requiring real-time data processing. Similarly, [Eliwa et al. \(2024\)](#) leverages the ANFIS regression model for predicting gasoline prices, showcasing how regression models can effectively handle noisy data in predictive analysis. [Farghaly and Abd El-Hafeez \(2023\)](#) proposes a high-quality feature selection method based on frequent and correlated items for text classification. This method could be adapted to improve event sequence analysis in DES by focusing on selecting the most relevant features for fault detection. Finally, [Farghaly and El-Hafeez \(2022\)](#) introduces a novel feature selection method based on frequent and associated item sets for text classification, which can also be applied to optimize the detection of critical sequences in DES.

The increasing complexity of Discrete Event Systems (DES), particularly in industrial contexts, necessitates robust and reliable fault diagnosis mechanisms ([Lefebvre et al., 2023](#)). Industrial systems are characterized by sequences of discrete events where faults or abnormalities can have significant consequences on safety, reliability, and efficiency ([Kampa et al., 2017](#); [Ignatiev et al., 2019](#)). Traditional models, such as Petri nets ([Zhou and Twiss, 1998](#); [David, 1995](#)) and automata ([Chesñevar et al., 2003](#)), have been extensively used for fault detection in DES, yet they face critical limitations, particularly in their ability to handle simultaneous events, reduce false positives, and recognize partial event sequences. These limitations hinder their effectiveness in real-time industrial processes, where precision and prompt fault detection are paramount ([Vásquez, 2017](#)). In response to these challenges, this paper introduces the formal theory of V-nets, a novel formal model designed to enhance the diagnosis and supervision of DES ([Vasquez-Capacho, 2020](#); [Vasquez Capacho et al., 2020](#)). V-nets build on the strengths of traditional models while addressing their shortcomings ([Niggemann and Lohweg, 2015](#)). They offer a flexible framework capable of managing simultaneous occurrences, minimizing the risk of false positives, and improving the system's ability to detect partial sequences. This adaptability makes V-nets particularly suited to industrial applications, where the complexity of event interactions and the need for accurate fault detection are critical. To demonstrate the practical utility of V-nets, this study presents a case analysis of the Reactor Startup Sequence in a Pressurized Water Reactor (PWR)—a highly complex and safety-critical process. In this case study, V-nets are applied to manage and diagnose fault sequences, showcasing their capability to enhance system monitoring and fault diagnosis. Compared to traditional models, V-nets provide improved flexibility and accuracy in identifying critical event sequences while reducing false positives. The integration of V-nets with automata further strengthens the model's ability to provide real-time supervision and diagnosis, making it a valuable tool for modern industrial systems.

This paper is organized as follows: Section 2 introduces the state-of-the-art and the theoretical foundations of V-nets, Section 3 discusses key parameters and characteristics; and Section 4 presents the Reactor Startup Sequence case study, illustrating the application of V-nets in a real-world industrial scenario. Section 5 presents a result analysis in which the process and algorithms used is explained. Finally Section 6 concludes the paper, highlighting the main findings and outlining directions for future research. By addressing the limitations of existing models and providing a comprehensive fault diagnosis solution, this study positions V-nets as a promising framework for improving the reliability and safety of complex industrial systems.

2. State of the art and foundational theory

The field of fault diagnosis in Discrete Event Systems (DES) has witnessed significant advancements, driven by the need for reliability and safety in industrial processes. Researchers have introduced various models and methodologies to tackle critical challenges, including simultaneous event occurrences, false positives in fault detection, and the

partial recognition of event sequences. This section provides a comparative analysis of recent developments in fault diagnosis for DES, with a focus on the methodologies employed, key contributions, and their effectiveness in addressing these challenges. While notable progress has been made, the persistent issues of handling event simultaneity and minimizing false positives continue to warrant further research and innovation. Table 1 presents a comparative analysis of key works in this field, summarizing their methodologies and identifying gaps that future research must address.

The comparative analysis of recent works in fault diagnosis for DES underscores the considerable advancements made in this field. Various methods have been proposed to enhance system performance, fault detection in partially observed systems, and data-driven fault diagnosis. However, several critical challenges remain underexplored. The simultaneous occurrence of events and the reduction of false positives in fault detection are still areas that require more robust and comprehensive models. In particular, the existing approaches tend to focus on improving specific aspects of fault diagnosis, such as efficiency in supervisory control or state estimation, but do not sufficiently address the complexities introduced by the simultaneity of events in industrial applications. Moreover, minimizing false positives in fault detection remains a challenge that has not been thoroughly solved. While fault diagnosis in Discrete Event Systems has progressed significantly, critical gaps remain in the ability of current models to handle simultaneous event occurrences and to reduce false positives. Addressing these challenges is paramount for ensuring the reliability and safety of industrial processes. Future research must focus on filling these gaps to develop more robust and reliable fault diagnosis systems.

On the other hand, V-nets provide a fresh theoretical framework for modeling and analyzing complex systems, particularly within the realm of discrete event systems (DES). This formalism is crafted to tackle challenges present in conventional methods, such as concurrent events, false positives, and partial recognition of event sequences. The theoretical foundation of V-nets establishes a robust and flexible system for representing and managing discrete events in dynamic processes. A V-net is fundamentally defined as a tuple comprising sets of events, temporal constraints, and a directed graph. The directed graph illustrates the connections between events and their temporal constraints. V-nets take a more adaptable approach to modeling dynamic processes, offering advantages over traditional state machines by avoiding the constraints of explicit state representations. The formal definition incorporates initiation and termination events, event occurrence frequencies, and logical predicates for validating V-nets' recognition. This approach brings a fresh perspective to event sequences by enabling the modeling of concurrent events, partial recognition, and tracking the frequency of events in temporal sequences. By clarifying these definitions, the goal is to enhance understanding and illuminate the practical applications of the V-net formalism. These core elements lay the groundwork for a deeper comprehension of how V-nets can be effectively applied across various scenarios.

Definition 1. In discrete event systems, a label designating a noteworthy event usually represents a *event type*. Let the set of all potential event types be represented by E . Within a system, an event type might stand in for a particular observation, action, or state change.

Example. Consider $a, b, c \in E$ representing event types such as process initiation, task completion, or specific conditions. Additionally, an *event type* can also be a logical combination of two or more other events. For instance, the event type $A = a.b$ signifies that event A occurs when both events a and b have transpired.

A mathematical theory known as “Time-stamped Observations Theory” (TOT) is structured to create artificial cognitive agents known as abstractors. These agents use an Abstraction Problem Solving Method ([Le Goc, 2006](#)) to analyze an input flow of timed messages and generate

Table 1

Comparative analysis of recent works in fault diagnosis for discrete event systems.

Work	Key contribution	Key challenges addressed
Wane and Charoy (2024) Optimal supervisory control of DES for cyclic tasks	Proposes a supervisory control framework optimized for cyclic tasks.	Improves system performance but does not focus on simultaneous events or false positives.
Qin et al. (2023) A Unified Framework for Verification of Observational Properties for Partially-Observed DES	Introduces a unified framework for verifying observational properties in partially observed DES, enhancing fault detection.	Addresses partial recognition but overlooks simultaneous event occurrences.
Zhang et al. (2023) Fault Detection and Fault-Tolerant Control for Discrete-Time Multiagent Systems	Data-driven fault detection framework tailored for multiagent systems with sensor faults.	Effective in handling sensor faults but lacks emphasis on event simultaneity.
Tian et al. (2023) Data-driven fault diagnosis approaches for industrial equipment: A review	Comprehensive review of fault diagnosis methods, with an emphasis on industrial applications.	Offers an extensive review but fails to provide specific solutions for simultaneous event occurrences.
Giua et al. (2024) Signal-Interpreted Colored Petri Nets for Discrete Event Systems	Introduces Petri nets for feedback control in DES, facilitating rapid prototyping.	Petri nets enhance sequence recognition but do not address simultaneous event handling.
Lin et al. (2023) Modeling and state estimation for supervisory control of networked timed DES	Focuses on state estimation in supervisory control of networked systems.	Provides effective state estimation but lacks solutions for simultaneous event occurrences.
Li et al. (2024) A uniform approach to compare architectures in decentralized DES	Compares decentralized DES architectures, enhancing system control and diagnosis.	Lacks focus on simultaneity and false positive reduction.
Piazzzi et al. (2007) Online Fault Diagnosis of DES: A Petri Net-Based Approach	Utilizes Petri nets for online fault diagnosis in DES.	Strong in online detection but does not address simultaneous events.
Sefrioui and Boucher (2013) Overview of fault diagnosis methods for DES	Provides an overview of diverse fault diagnosis techniques for DES.	Comprehensive review but lacks detailed solutions for handling simultaneous events.
Ding et al. (2019) Robust Fault Diagnosis of Stochastic DES	Proposes robust fault diagnosis techniques for stochastic DES.	Effective for stochastic systems but overlooks event simultaneity issues.

a minimal but relevant output flow of timed observations. Le Goc's Theory of Timed Observation applies Shannon's Theory of Communication to timed data and provides a distinct framework for Poisson Theories and Markov Chains. A process $Pr(t) = \{x_1(t), x_2(t), \dots, x_n(t)\}$ of time functions $(x_i(t))$ is considered to be a dynamic system. A program Θ monitors the evolution of these time functions, and $X = \{x_1, x_2, \dots, x_n\}$ defines the collection of variable names. For each $\Delta_{x_i} = \{\delta_1^i, \delta_2^i, \dots, \delta_m^i\}$, a collection of constants representing the potential values for x_i is constructed as follows: $\Delta = \bigcup_{x_i \in X} \Delta_{x_i}$. Consider the collection of random time instants $\Gamma = \{t_k\}_{t_k \in \mathbb{R}}$ (Pomponio and Goc, 2014; Curt et al., 2017).

Definition 2 & 3 (Timed Observation and Event Type Occurrence).

When a program Θ observes a time function $x_i(t)$ at time $t_k \in \Gamma$, it makes a timed observation $o(t_k) \equiv (\delta_j^i, t_k)$. This is achieved by assigning the values $v = x_i$, $\delta v = \delta_j^i$, and $t = t_k$ to a predicate $\Theta(v, \delta v, t)$ so that: $\Theta(x_i, \delta_j^i, t_k)$ (Pomponio and Goc, 2014; Curt et al. (2017)). An event type occurrence, also known as an event e , is defined as an event type occurrence pair $e = (\sigma, t)$, where t is an integer variable that represents the event date and $\sigma \in E$ indicates the event type. Any change in the parameters or values of a given domain feature or group of features is indicated by each event type σ . The ordered set that includes every sort of event involved in a particular process or industrial application is defined as E in this context. These occurrences also line up with the values δ_j^i that have been tracked over time. In other words, when we talk about an event or a time observation, we are essentially talking about a date of occurrence.

A sequence or trace is a flow of actions produced by a system, where time is represented by a discrete collection of time points. It is presumed that points are correct if they are entirely ordered and have a granularity narrow enough relative to the dynamics that have been observed and the possible accuracy that may be derived from observation. Thus, a temporal sequence could represent how a system operates. An orderly sequence of events is known as a temporal sequence or just a sequence for short. This definition leads to the following:

Definition 4. An ordered set of timed observations on E is called a sequence or trace. $S = \langle (t_k, \delta_j^i) \rangle_m$, where l is the length of the temporal sequence and $m \in \mathbb{N}_l$. A finite set of linearly ordered instants of cardinal l is represented by S and N_l . The number of timed observations in S , or

the size of the temporal sequence, is denoted by $l_{(S)} = |S| \cdot t_i < (t_{i+1})$ and $i = 1 \dots l$, $i = 1 \dots l - 1$, where l denotes the duration of the time series S . A sequence can serve as an example of a trace or sequence that represents an activity stream. $o(t_1), o(t_2), o(t_3), o(t_4), o(t_5) = \{(b, 3), (c, 5), (a, 6), (b, 9)\}$ with $l_{(S_x)} = 5$.

Definition 5. When two or more event categories occur simultaneously or at such close intervals that they can be called simultaneous, the scenario is referred to as a simultaneous occurrence. In this notion, simultaneous occurrences are represented by a subset of event types, $E^* \subset E$. Consider a process having event types a, b , and c , for example. These events can occur concurrently in some cases.

Definition 6. The sequences of characteristic or learning event sequences of a given situation are represented by the sequence of event types, S^r . Offline, these sequences can be used as a guide to create temporal patterns.

Definition 7. The event type sequence, represented by the notation S^e , stands for the evaluation sequences that are utilized to validate a temporal pattern online.

Definition 8. σ^f is the definition of the frequency for each event type σ in a single sequence S_n . For instance, a^2 can be used to represent the event type a when a timed observation of it occurs twice.

For any kind of event, the date of occurrence is defined as t^{fn} , where fn denotes the number of times this kind of event had occurred. For instance, the date of the third instance of the kind of event a is represented as $t^3 a$. Furthermore, $f^{max}(\sigma)$ defines the maximal frequency of occurrence for an event type in a temporal sequence.

V-nets are a novel formalism that was developed especially for modeling and studying complex systems that evolve and are characterized by discrete occurrences. Vasquez Capacho et al. (2020) and Vasquez-Capacho (2020) presented this formalism in their seminal publications. In contrast to conventional Petri nets, V-nets use squares rather than circles to symbolize events, doing away with the need for locations. The arcs that join these squares take into account both frequency occurrences and time limits. Alongside this formalism is a collection

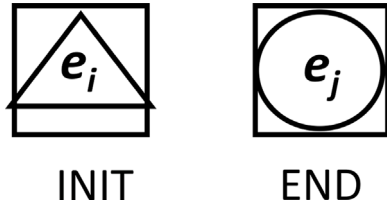


Fig. 1. Graphical representation of INIT and END.

of attributes that characterize the system's behavior, such as transition firing rules, simultaneous event handling standards, and false positive analysis techniques. Compared to previous formalisms, V-nets provide a more accurate and effective modeling technique for complex systems and fault diagnostics, especially in the areas of Cyber-Physical Systems and industrial applications. In addition, V-nets show promise for industrial use in efficiency diagnosis by giving a graphical depiction of discrete events and how they relate to one another in the system. A V-net is precisely defined as follows.

Definition 9. A V-net (V_N) is defined as the tuple $\langle E, \mathcal{T}, \mathcal{G}, \text{INIT}, \text{END}, \text{Frec}, tl_{eval}, \mathbf{R} \rangle$ such that:

- E is the set of event types involved in the V_N ,
- $\mathcal{T}$ is the set of temporal constraints of the V-net,
- $\mathcal{G} = (\mathcal{E}, \mathcal{A}, \text{Ev})$ is a directed graph in which:
 - $\mathcal{E}$ is the set of the graphical representations in squares of the event types involved in the V_N ,
 - Arcs $\mathcal{A}$ represents the different time constraints between events.
 - $\text{Ev}:(\text{True}/\text{false})$ represents the activation or deactivation function of the arcs $\mathcal{A}$ according to the evolution of the timed observations.
- **INIT** corresponds to the event or events that initiate the event sequence, see Fig. 1
- **END** corresponds to the event or events that end the event sequence, see Fig. 1. In a V-net, the initial event and the final event could be the same.
- **Frec** corresponds to the frequencies for each event in the event sequence. For example, in the interval or temporal constraint $e_i^{f_{e_i}}[I^-, I^+]^{f_{e_j}}e_j$, where $f_{e_i}(e_j)$ corresponds to the frequency of the event e_i (e_j) on the interval $[I^-, I^+]$.
- tl_{eval} corresponds to the time of evaluation of the V-net after the occurrence of its first observation.
- $\mathbf{R}$ contains the set of logical predicates that correspond to the restrictions and warnings that confirm the V-net recognition.

Examples:

- **Frec**(a):5
- **Frec**(a):1 $\wedge$ **Frec**(b):1
- **Warning**: $a = b$
- **Warning**: Additional b after a^2

Example 1 (Single V-net). Consider the V-net defined by

- $E = \{a, b\}$
- $\mathcal{T} = \{t_1(a^1 \Rightarrow b^1) : [1, 2]\}$
- **Frec** = $\{f^{max}(a) : 1; f^{max}(b) : 1\}$
- $tl_{eval} : 2$
- $\mathbf{R} = \text{Frec}(b) : 1$

In this simple V-net, the event a is the first event that occurs (INIT), and after between 1 and 2-time units, the event b (END) must occur, and

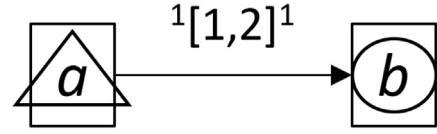
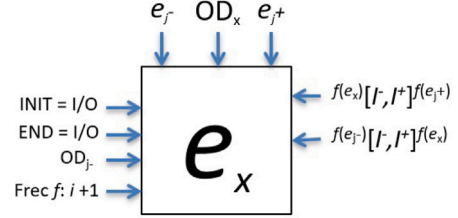


Fig. 2. Example 1. Single V-net.

Fig. 3. Representation of the event e_x as an object.

then this V-net is recognized. The graph $\mathcal{G}$ for this V-net is presented in Fig. 2

Definition 10. Regarding the assessment of V-nets, a V-net event can be regarded as an entity with characteristics and parameters that change based on the dynamics of the event occurrences in the evaluation sequence S^e . An illustration of this object is shown in Fig. 3.

Let us assume that e_x signifies the occurrence of an event type within an event sequence of evaluation S^e .

- OD_x corresponds to the date of occurrences of e_x
- e_{j-} corresponds to the event type that had occurred before of e_x , this not exists if e_x is represented as INIT.
- e_{j+} corresponds to the event type that must occur after of e_x according to the V-net, this will not exists if e_x is represented as END
- INIT is a mark of 1 or 0 depending on the representation of e_x in the graph
- END is a mark of 1 or 0 depending on the representation of e_x in the graph
- OD_{j-} is the date of e_{j-} when it had occurred just before of e_x
- **Frec** $f: i+1$ corresponds to the frequency of times that event type e_x had occurred. This counter i initiates in zero and goes incrementing each time e_x occurs.
- According to the V-net graph, $f_{(e_x)}[I^-, I^+]^{f_{(e_{j+})}}$ corresponds to the time transition between e_x and the next event type occurrence.
- According the V-net graph, $f_{(e_{j-})}[I^-, I^+]^{f_{(e_x)}}$ corresponds to the time transition between e_x and event type that had occurred before.

3. Important variables and traits

Understanding system behavior and dynamics in V-nets depends critically on certain parameters and features. For instance, time liveness describes how long a V-net runs and gives information on the temporal characteristics of the system. Moreover, V-net dynamics capture the way the simulated system changes over time, including state transitions, transitions firing, and general system behavior. Finally, event sequence analysis in V-nets examines trends and time correlations between events to help identify abnormalities and repeating sequences. The following variables work together to provide a thorough insight into system dynamics and are essential for maximizing efficiency and identifying problems in intricate systems.

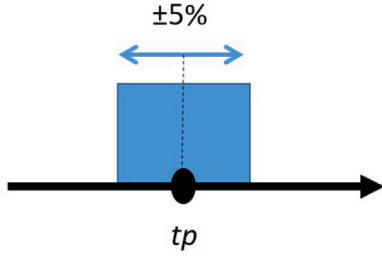
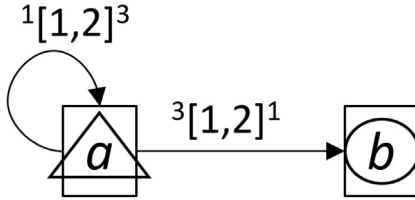


Fig. 4. Example of bandwidth limits.

Fig. 5. Example 2 of a V-net (V_N).

3.1. Limitations on bandwidth

The value of the time units is determined by the level of granularity required in the system, and the time points for the analysis of the event sequences correspond to a finite collection of linearly ordered instants. The necessary precision and variability can be used to set the bandwidth limits for each time point. For all event-type occurrences inside this bandwidth, deviations of 5% before and 5% after the time point, for instance, may be allowed. A bandwidth limit example is shown in Fig. 4, where the time point tp has two limits: low $tp - \%5$ and high $tp + \%5$. This can be useful in calculating the analysis of simultaneous events and analyzing the date of each event that occurs close to a time point.

3.2. Timed language

The alphabet in the V-nets (V_N) correlates to E , in line with language theory. The strings w that can be formed based on the arrangement of the events on the graph will reference the timed language $L^T(E, \mathcal{T})$ of each V_N . The set of temporal constraints $\mathcal{T}$ will limit the V-net and the concatenation of the components in the strings.

Example 2 (V-Net with 2 Arcs).

Example 2 of V-net presented in Fig. 5 in this case contains:

- $E = \{a, b\}$
- $\mathcal{T} = \{t_1(a^1 \Rightarrow a^3) : [1, 2]; t_2(a^3 \Rightarrow b^1) : [1, 2]\}$
- $\mathbf{Frec} = \{f_a^{max}(a) : 3; f_b^{max}(b) : 1\}$
- $tl_{eval} : 6$
- $R = \mathbf{Frec}(b) : 1$

3.3. Event sequences that V-nets can identify

There must be a finite number of timed observations in an event sequence S . $S = \langle (e_i, t_i) \rangle$, where $l_{(S)} = |S| = n$ and $j \in N_I$ is a finite set of linearly ordered time points. $\bigcap_{e_i \in S} e = \{e | e \in \bigcap_{j=1}^n e_i\}$ is the integration of all the events e_i included in S , corresponding to all the events and each from event e_1 to event e_n , where n is a finite number. The sample sequence of events S_1 (see Fig. 6) contains $l_{(S_1)} = 4$ timed occurrences and a finite number of events $|E'| = 2$ (a and b). $L^T = \{(a, t_1), (a, t_2), (a, t_3), (b, t_4)\}$ is the timed language, where $t_1 < t_2 < t_3 < t_4$. As a result, this series of events (S_1) can meet the deadlines and the required quantity of occurrences for V-net Example 2.

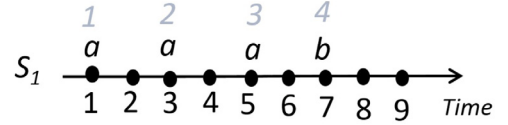
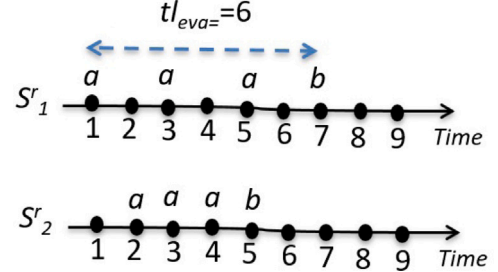


Fig. 6. Example of an event sequence.

Fig. 7. S^r for Example 2 of V-net and its maximal time length.

3.4. Frequencies of events in a single series of events

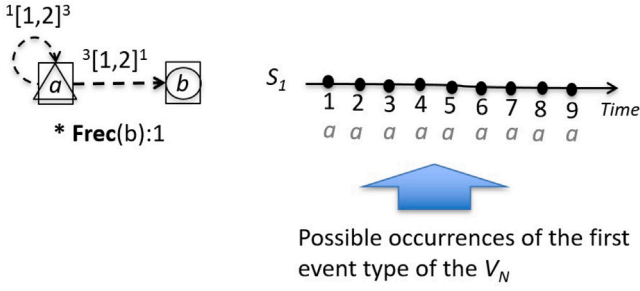
An event type σ has a finite frequency in a single event sequence. $\mathbf{Frec}(\sigma) \leq f_a^{max}$ is the maximal frequency of an event type σ if f_a^{max} is the maximum number of timed occurrences in a sequence. In Fig. 5, we have $f_a^{max} = 3$ and $f_b^{max} = 1$ for the example. When an event frequency appears in parenthesis during the V-net evaluation process, it means that the time observation does not match the frequency specified on the V-net's time restriction. After the evaluation, a warning appears to show that the valid event occurred before the abnormal time observation that was obtained. As an illustration, consider the time constraints $a^3[1,2]^1b$ and **Warning:** Additional b after a^2 indicates that one additional occurrence of the event type b happened after the second occurrence of the event type a .

3.5. Types of events in the directed graph

There is just one representation of each event in the directed graph G . The first event e_m :(INIT) and the last event e_k :(END) exist in the directed graph from all the representative sequences of events S^r , or in other words, the event types described from the automata that handle the control of the system. More initial and final events could occur on the V-net, as shown by the following: $\forall S^r \Rightarrow \exists \{e_m : INIT\}, \{e_k : END\}$. The directed graph in the V-net example (see Fig. 5) depicts the events a : INIT and b : END as they appear in the representative event sequences (see Fig. 7).

3.6. V-net's time of existence

The amount of time that passes between the first and last timed observation of an event sequence is called its time length (tl). For instance, the time length for a sequence S is $tl = (t_n - t_i)$ if the first and last timed observations in the series are (e_i, t_i) and (e_n, t_n) , respectively. The longest of them will now be the time duration indicated tl_{eva} used for the evaluation of any event sequence S in the V-net, derived from the collection of representative event sequences S^r or from the analysis of the automata governing the control of the system. Following the initial timed observation, the V-net will only accept event occurrences that occur before the end of the assessment period. Fig. 7 indicates that the maximum time length of the representative event sequences (S^r_1 and S^r_2) is $tl_{eva} = 6$.

Fig. 8. Possible timed observations to this V_N .

3.7. Dynamics of V-nets

The evolution of the timed observations in the directed graph $\mathcal{G}$ is activated if only congruence is present between the frequency of the event and the frequency indicated in the time interval. **Example 2** of the V-net will be evaluated with the evaluation event sequence S_1^e , in which the evolution of the time observation in this V-net is presented from Figs. 8 to 22.

Fig. 8 presents the possible timed observations to this V_N of **Example 2**; let us check that the arcs are not activated yet because no event type occurrence has occurred and the first event that must happen is a .

Fig. 9 presents the first timed observation to this V_N of **Example 2** ($a,2$); let us check that one arc is active waiting for the next event type occurrence.

Fig. 10 presents the graphical representation of V – net event: a^1 .

Fig. 11 presents the possible second timed observation to this V_N of **Example 2**.

Fig. 12 presents the second timed observation to this V_N of **Example 2** ($a,2$) satisfying its temporal restriction; let us check that the arc is active and waiting for the next (third) event-type occurrence.

Fig. 13 presents the graphical representation of V – net event: a^2 .

Fig. 14 presents the possible third timed observation of this V_N of **Example 2**.

Fig. 15 presents the third timed observation to this V_N of **Example 2** ($b,5$), which does not satisfy its temporal restriction because the arc that connects with b is not active yet. In this case, the frequency of b is in parentheses.

Fig. 16 presents the graphical representation of V – net event: b^1 . Note that the frequency is assumed as (1) because the V-net is waiting for an event type a not for a b . In addition, this time occurrence does not have time restrictions available.

Fig. 17 presents the possible fourth timed observation to this V_N of **Example 2**, it is waiting for the last a .

Fig. 18 presents the fourth timed observation on this V_N of **Example 2** ($a,4$), which satisfies its temporal restriction.

Fig. 19 presents the graphical representation of V – net event: a^3 .

Fig. 20 presents the possible fifth timed observation of this V_N of **Example 2**, note that a new arc is active, and at this moment it is waiting for END with b .

Fig. 21 presents the fifth timed observation on this V_N of **Example 2** ($b,8$), this is the final event that recognizes the V-net, and restarts it waiting for a new event sequence.

Fig. 22 presents the graphical representation of V – net event: b^2 .

3.8. Connection of the events on graph $\mathcal{G}$

An event is connected with another event in $\mathcal{G}$ if just at least one incidence or precedence of them on the representative event sequences exists. There is an arc $\mathcal{A}$ between e_i and e_j if on the timed observations e_j is the successor of e_i or e_j is the predecessor of e_i .

Example 3 (V-net with 4 Events).

Example 3. is presented in Fig. 23, where is indicated its four representative event sequences (S_1^r , S_2^r , S_3^r and S_4^r), and this V-net is defined by.

- $E = \{a, b, c, g\}$
- $\mathcal{T} = \{t_1(a^1 \Rightarrow b^1) : [1, 1]; t_2(a^1 \Rightarrow c^1) : [1, 1]; t_3(b^1 \Rightarrow c^1) : [1, 1]; t_4(c^1 \Rightarrow b^1) : [1, 1]; t_5((c^1 \rightarrow b^1) \Rightarrow g^1) : [1, 2]; t_6((b^1 \rightarrow c^1) \Rightarrow g^1) : [1, 2]\}$
- $\mathbf{Frec} = \{f^{max}(a) : 1; f^{max}(b) : 1; f^{max}(c) : 1; f^{max}(g) : 1\}$
- $tl_{eval} : 4$
- $R = \mathbf{Frec}(g) : 1; \mathbf{Warning} : b = c$

In $\mathcal{G}$ the event a is connected to events b and c because in representative event sequences, b and c always follow, after that appearing the event a , and in addition, b and c never happen simultaneously. This is a special situation that generates a warning. The notation “ $b = c$ ” indicates that these events occur at the same time. Note that the expression $c^1 \rightarrow b^1$ in the time restriction t_5 expresses that activation of the event b^1 is possible only if the event c^1 had occurred before. Similar in t_6 with $b^1 \rightarrow c^1$ but in this case, the activation of the event c^1 only is possible if the event b^1 had occurred before.

3.9. Occurrence matrix

The occurrence matrix is a timeless representation of the event sequences that accept the V-net. The columns of the matrix represent the events of V_N , and the rows represent all ordered occurrences on the possible sequences of events, beginning with INIT and ending with END. This characteristic of the V_N allows for a preliminary review of all the possible order of the event occurrences permitted to evaluate the V-net. Inside these matrices, 1 represents the timed observations of the event types and 0 represents that the event type had not occurred.

Example 4 (V-Net with 3 Events and Its Occurrence Matrices).

Example 4. is presented in Fig. 24, where is indicated its three occurrence matrices, and this V-net is defined by.

- $E = \{a, b, c\}$
- $\mathcal{T} = \{t_1(a^1 \Rightarrow b^1) : [1, 1]; t_2(a^1 \Rightarrow c^1) : [1, 1]; t_3(b^1 \Rightarrow c^1) : [0, 1]; t_4(c^1 \Rightarrow b^1) : [0, 1]\}$
- $\mathbf{Frec} = \{f^{max}(a) : 1; f^{max}(b) : 1; f^{max}(c) : 1\}$
- $tl_{eval} : 3$
- $R = \mathbf{Frec}(b) : 1 \wedge \mathbf{Frec}(c) : 1$

The V-net example depicted in Fig. 24 showcases three potential occurrence matrices. In the first matrix, the initial event (INIT) is labeled a , followed by the simultaneous occurrence of events b and c (END). It is important to note that in this instance, simultaneous occurrences of b and c are permissible, and a specific designation of “simultaneous occurrence” is not necessary to denote this condition. In the second matrix, the sequence begins with event a (INIT), followed by event b (Oc1), and concludes with event c (END). Finally, the third matrix illustrates another scenario where the sequence starts with event a (INIT), proceeds with event c (Oc1), and concludes with event b (END). Note that in the time restrictions from b to c and from c to b it is permitted simultaneous occurrences due to the value of zero 0.

3.10. Partial recognition

Definition 11. V-net instance. A V-net $\langle E, \mathcal{T}, \mathcal{G}, \mathbf{INIT}, \mathbf{END}, \mathbf{Frec}, tl_{eval}, R \rangle$ is recognized when occurs a temporal sequence S of events E' , such that $\xi \subseteq E'$ and it satisfies all temporal constraints $\mathcal{T}$.

A V-net is partially recognized (%Rec) when occurs a temporal sequence S of events E' , such that $|E'| = |\xi|$ and it partially satisfies all temporal constraints $\mathcal{T}$. Also, it is partially recognized, if the quantity of the timed occurrences in S is less than the quantity indicated in

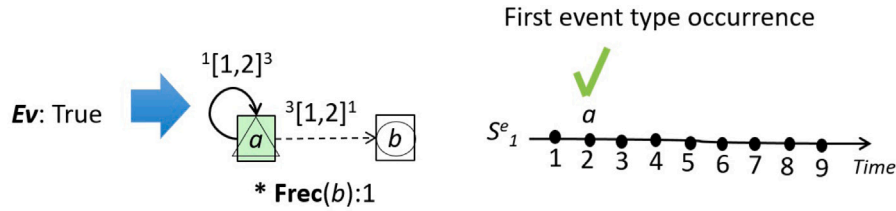
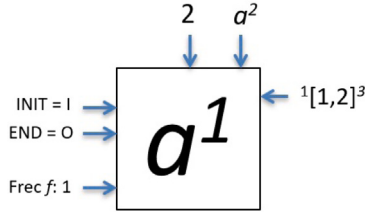


Fig. 9. First timed observation (a,2).

Fig. 10. Representation of the event a^1 as an object.

the V-net. This guarantees that the exact number of events that will occur will satisfy all temporal constraints. If there are additional event occurrences, although the V-net could be recognized in 100%, warnings may be generated for each additional timed observation that does not match the V-net.

Example 5 (V-Net with 4 Events and Its Partial Recognition).

Example 5. is presented in Fig. 25, where is indicated its partial recognition and this V-net is defined by.

- $E = \{a, b, c, d\}$
- $\mathcal{T} = \{t_1(a^1 \Rightarrow b^1) : [3, 4]; t_2(a^1 \Rightarrow c^1) : [0, 0]; t_3(b^1 \Rightarrow d^1) : [1, 2]; t_4(c^1 \Rightarrow b^1) : [3, 4]\}$
- $\mathbf{Frec} = \{f^{max}(a) : 1; f^{max}(b) : 1; f^{max}(c) : 1; f^{max}(d) : 1\}$
- $tl_{eval} : 6$
- $R = \mathbf{Frec}(d) : 1$

The example presented in Fig. 25, indicates that the sequence S_3^e recognized the V-net in 75%, because $\%Rec = (3/4) \cdot 100$.

$$\%Rec = \frac{\text{Number of events recognized}}{\text{Number of timed observations}} \cdot 100 \quad (1)$$

The example of V-net in Fig. 25 has the following set of events $E = \{a, b, c, d\}$, and S_1^e and S_2^e recognized 100% the V-net; on contrary S_3^e does not, because event c occurs after event a , and both must happen simultaneously.

4. Case study: Reactor startup sequence for a pressurized water reactor (PWR)

The launch sequence of a Pressurized Water Reactor (PWR) is a crucial phase in the operation of nuclear power generation plants, marking the transition from shutdown to full power production. This process entails a meticulously planned series of steps aimed at initiating and controlling the nuclear fission process within the reactor core while ensuring the safety and stability of the entire system. The PWR design, renowned for its widespread use in commercial nuclear power plants, relies on a closed-loop cooling system where pressurized water serves as both a coolant and a moderator. In this case study, we dive into the intricate details of the reactor startup sequence for a PWR, focusing on the initiation of the start procedures and the systematic activation of key reactor components. We explore the essential steps involved in control rod manipulation, neutron flux stabilization, coolant system

Table 2

Cold shutdown state: Process variables and transition conditions.

Primary component	Reactor core
Event Sequence	Initialization of reactor systems, pre-startup checks
Response of Process Variables	Temperature: Low, Pressure: Low, Neutron Flux: Absent
Transition Conditions	Pre-startup checks successfully completed. No anomalies detected.

activation, steam generation, turbine synchronization, and grid connection. Through a comprehensive examination of the startup sequence, we gain insights into the operational challenges, safety considerations, and regulatory compliance inherent in the startup process of a PWR. Additionally, we highlight the significance of leveraging advanced modeling and analysis techniques, such as V-nets, to enhance the supervision and monitoring of reactor startup procedures, thereby ensuring the safe and efficient operation of nuclear power generation plants.

4.1. Automata definition

Automata plays a crucial role in modeling and supervising the Reactor Startup Sequence for a Pressurized Water Reactor (PWR). These finite-state machines provide a structured framework for representing the sequential progression of events and states during the startup process, ensuring the safe and efficient initiation of reactor operations. In this detailed description, we will outline each step of the Reactor startup sequence, highlighting the primary industrial components involved, the event sequences required, and the response of process variables such as temperature, pressure, etc., categorized into high, medium, or low levels. An example of this automata is presented in Fig. 26 in which can be observed its five states with its four transition events

4.1.1. State 1: Cold shutdown

In this state, the reactor core is in a non-operational state with all reactor systems being initialized for safety. Table 2 summarizes the key process variables, event sequences, and transition conditions involved in the Cold Shutdown state.

4.1.2. State 2: Initiation of reactor startup

This state involves the activation of the reactor through the withdrawal of control rods and initial fission reaction initiation. Table 3 outlines the event sequence, process variable responses, and transition conditions in the Reactor Startup state.

4.1.3. State 3: Power ascension

During the Power Ascension state, the reactor power is gradually increased. Table 4 summarizes the primary events, process variable responses, and transition conditions for this state.

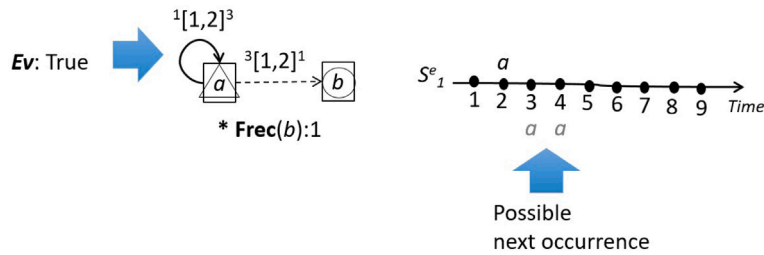


Fig. 11. Possible second timed observation.

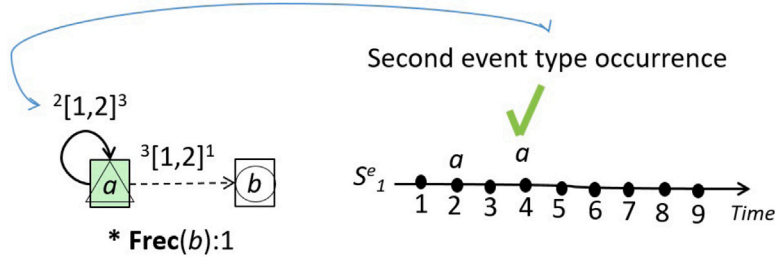


Fig. 12. Second timed observation (a,4).

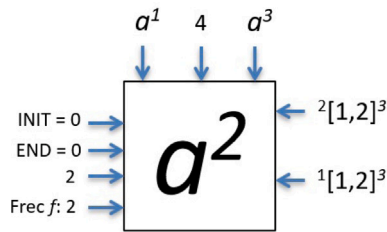
Fig. 13. Representation of the event a^2 as an object.

Table 3

Initiation of reactor startup: Process variables and transition conditions.

Primary component	Control rods
Event Sequence	Control rod withdrawal, neutron flux measurement, coolant system activation
Response of Process Variables	Temperature: Medium, Pressure: Medium, Neutron Flux: Medium
Transition Conditions	Neutron flux stabilization, coolant temperature and pressure reaching operational levels

Table 4

Power ascension: Process variables and transition conditions.

Primary components	Reactor core and turbine
Event Sequence	Increase in reactor power output, steam generation, turbine activation
Response of Process Variables	Temperature: High, Pressure: High, Neutron Flux: Medium to High
Transition Conditions	Stable reactor power output, steam supply to turbine within specified parameters

Table 5

Full power operation: Process variables and transition conditions.

Primary component	Power grid connection
Event Sequence	Grid connection, full power generation, continuous monitoring
Response of Process Variables	Temperature: High, Pressure: High, Neutron Flux: High
Transition Conditions	Successful grid synchronization, sustained full power generation

Table 6

Normal operation: Process variables and transition conditions.

Primary components	Reactor core and safety systems
Event Sequence	Routine maintenance, periodic inspections, emergency response preparation
Response of Process Variables	Temperature: Medium, Pressure: Medium, Neutron Flux: Medium
Transition Conditions	No anomalies detected, operational emergency response systems

4.1.5. State 5: Normal operation

In the normal operation state, the reactor maintains regular functionality with safety protocols actively monitored. Table 6 outlines the main events, process variables, and conditions for transitioning to other states from normal operation.

4.2. V-net construction

This section delves into the construction of V-nets tailored specifically for the Reactor Startup Sequence of a Pressurized Water Reactor (PWR). V-nets, a novel formalism for modeling and analyzing dynamic systems characterized by discrete events, offer a powerful toolset for capturing the intricate dynamics of complex industrial processes. The Reactor Startup Sequence of a PWR represents a critical phase in the operation of nuclear power plants, encompassing a series of meticulously orchestrated steps aimed at safely initiating and ramping up reactor power output. Each stage of this sequence is marked by discrete events, ranging from the initiation of reactor start procedures to the achievement of full power operation and subsequent transition to normal operational mode. To construct effective V-nets for the Reactor

4.1.4. State 4: Full power operation

The Full Power Operation state marks the peak performance of the reactor with stable power generation. Table 5 describes the process variables, event sequences, and conditions for transitioning from this state.

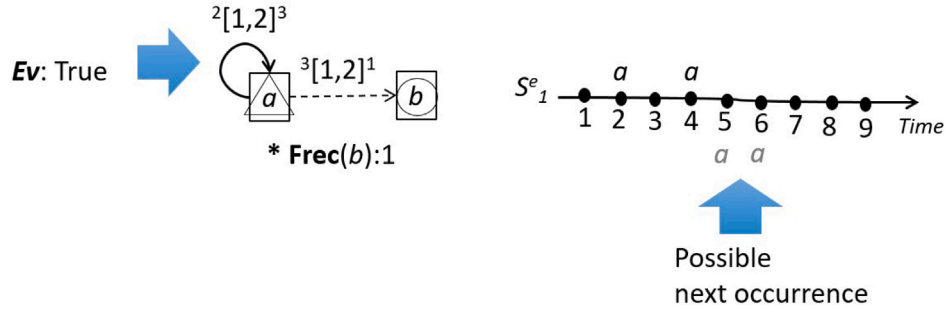


Fig. 14. Possible third timed observation.

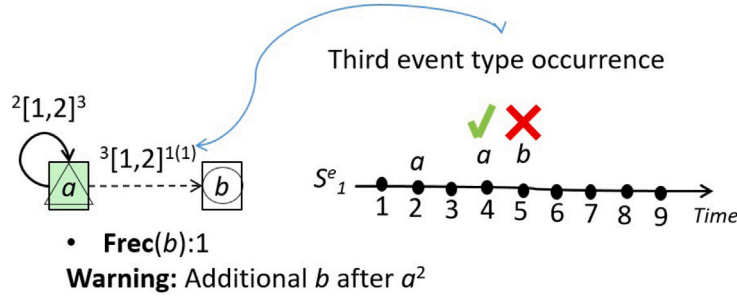
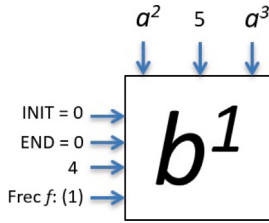


Fig. 15. Third timed observation (b,5).

Fig. 16. Representation of the event b^1 as an object.

startup sequence, researchers must first identify and define the discrete events associated with each stage of the process. These events are informed by the expertise of nuclear engineers, plant operators, and safety specialists who have a deep understanding of the operational nuances and safety protocols inherent to nuclear power plant operations. Furthermore, researchers will explore two or more plausible event sequences representative of typical operational scenarios encountered during the Reactor Startup Sequence. These event sequences serve as the building blocks for V-nets, providing concrete examples of how different combinations of events unfold over time and influence the behavior of the system. Through collaborative efforts between domain experts and modeling specialists, the aim is to construct V-nets that not only accurately represent the Reactor Startup Sequence but also offer valuable insights into system dynamics, performance metrics, and potential areas of improvement. Using the rich contextual information provided by experts, V-nets will serve as invaluable tools to enhance the safety, efficiency, and reliability of nuclear power plant operations.

4.2.1. V-net for State 1

The Cold Shutdown phase typically unfolds with the following event sequence, each event occurring within specified time restrictions:

Event a (Initialization of Reactor Systems) occurs first, initializing the primary and secondary coolant systems. Following Event a , Event A (Pre-Startup Checks) takes place, involving safety checks and system verification. This event should occur within a time frame of 5 to 10 min after the start of the shutdown phase. These time restrictions ensure

that the reactor systems are initialized promptly and that safety checks are conducted without undue delay, laying the groundwork for the subsequent stages of the reactor startup process. On the other hand, the discrete events defined for the variables will be:

- Temperature
 - Low: T_l
 - Medium: T_m
 - High: T_h
- Pressure
 - Low: P_l
 - Medium: P_m
 - High: P_h
- Neutron Flux
 - Absent: N_l
 - Medium: N_m
 - High: N_h

In this case, the event A represents the following expression $A: T_l.P_l.N_l$ which indicates that at the activation of the event A , the evaluation of the process variables will determine that the events T_l , P_l , and N_l must be active representing the correct value of the process variables.

Fig. 27 presents the representative sequences of events in this state. Furthermore, Fig. 28 presents the graphical representation of the V-net to supervise this State 1. Note that the event A is the final event of the V-net and the transition event to the next state.

V_{N1} is defined by

- $E = \{a, A\}$
- $\mathcal{T} = \{t_1(a^1 \Rightarrow A^1) : [10, 15]\}$
- $\mathbf{Frec} = \{f^{max}(a) : 1; f^{max}(A) : 1\}$
- $t_{eval} : 15$
- $R = \mathbf{Frec}(A) : 1, A = T_l \wedge P_l \wedge N_l$

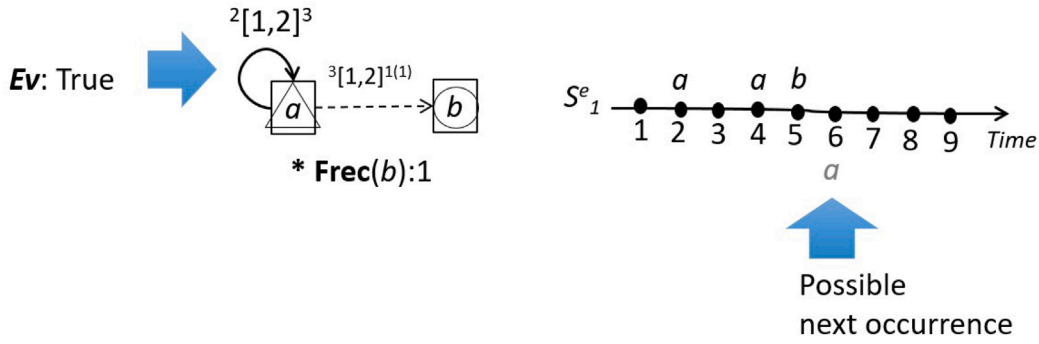


Fig. 17. Possible fourth timed observation.

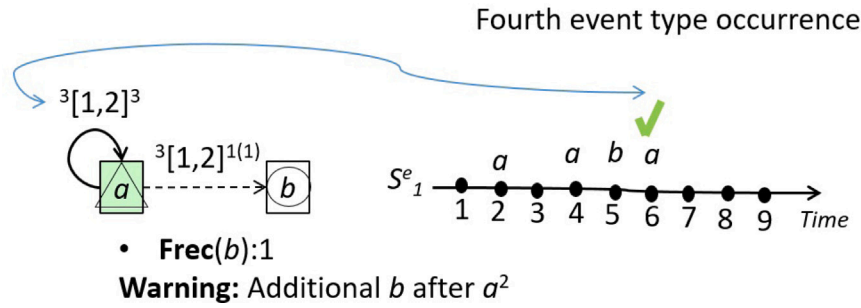
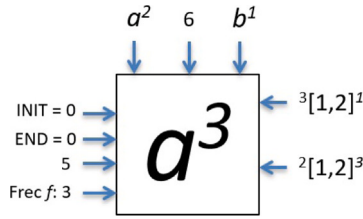


Fig. 18. Fourth timed observation (a,6).

Fig. 19. Representation of the event a^3 as an object.

The condition $A = T_l \wedge P_l \wedge N_l$ must be arranged after a between 10 and 15 min.

Fig. 28 illustrates the concept of V-nets as diagnostic tools used for monitoring event sequences within specific states of a process. Each V-net depicted in the figure corresponds to a particular state of the system being monitored, meaning that for every state, there is a unique V-net that captures the relevant events and their relationships. The primary function of these V-nets is to serve as diagnostic tools that monitor the sequences of events expected to occur within each state. By doing so, V-nets help ensure that the process operates as intended and that all necessary events are executed in the correct order. Within each V-net, the event sequences represent the specific actions or observations that must take place in that state, which are crucial for maintaining the integrity of the process. If the expected events do not occur, or if they occur out of order, the V-net can identify these discrepancies. The V-nets continuously monitor the occurrence of events in real-time, allowing them to detect simultaneous events, false positives, and partial recognition. For instance, if multiple events are supposed to occur at the same time, the V-net can recognize this and ensure that the process is functioning correctly. Additionally, the V-net can help identify when an event is incorrectly recognized, enabling adjustments to be made in the monitoring system. If only part of the expected event sequence occurs, the V-net can flag this as an issue, prompting further investigation. The information gathered by the V-nets can be used to provide feedback

to system operators, informing them of any deviations from the expected event sequences. This feedback allows for timely interventions to correct issues before they escalate. The figure likely includes visual elements such as nodes representing events, directed edges showing the relationships between events, and possibly annotations indicating the expected sequences. This visual representation helps clarify how the V-nets function as diagnostic tools. In summary, Figure Fig. 28 effectively illustrates how V-nets are utilized as diagnostic tools for monitoring event sequences within specific states of a process. By capturing the expected sequences of events, V-nets enhance the ability to detect anomalies, ensure compliance with operational protocols, and ultimately improve the reliability and safety of the system being monitored, which is particularly valuable in complex systems where precise event management is critical.

4.2.2. V-net for State 2

During State 2: Initiation of the reactor startup, the startup sequence involves a series of operational actions and responses of process variables. The sequence begins with the Control Rod Withdrawal (CRW) (event b), where operators initiate the withdrawal of control rods to facilitate the increase in neutron flux. This action leads to a transition of neutron flux from low (N_l) to medium (N_m) levels between 10 and 15 min. Subsequently, the Coolant System Activation (CSA) (event c) occurs, where the operators activate the components of the coolant system to manage the reactor temperature. This action leads to the onset of an increase in temperature and pressure from their low levels (T_l , P_l). The activation of the coolant system should be completed within 20 to 30 min after the neutron flux measurement to maintain operational safety. These discrete events, marked by their respective operational actions and transitions in the process variables, ensure a systematic and controlled startup process for the pressurized water reactor (PWR). Fig. 29 presents the representative sequences of events in this state. Furthermore, Fig. 30 presents the graphical representation of the V-net to supervise this State 2. Note that in this case, event B is the final event of the V-net and the transition event to the next state.

V_{N2} is defined by

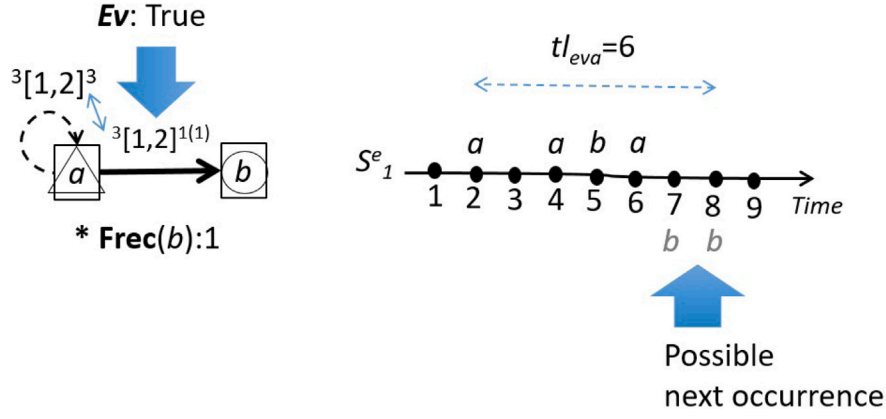


Fig. 20. Possible fifth timed observation.

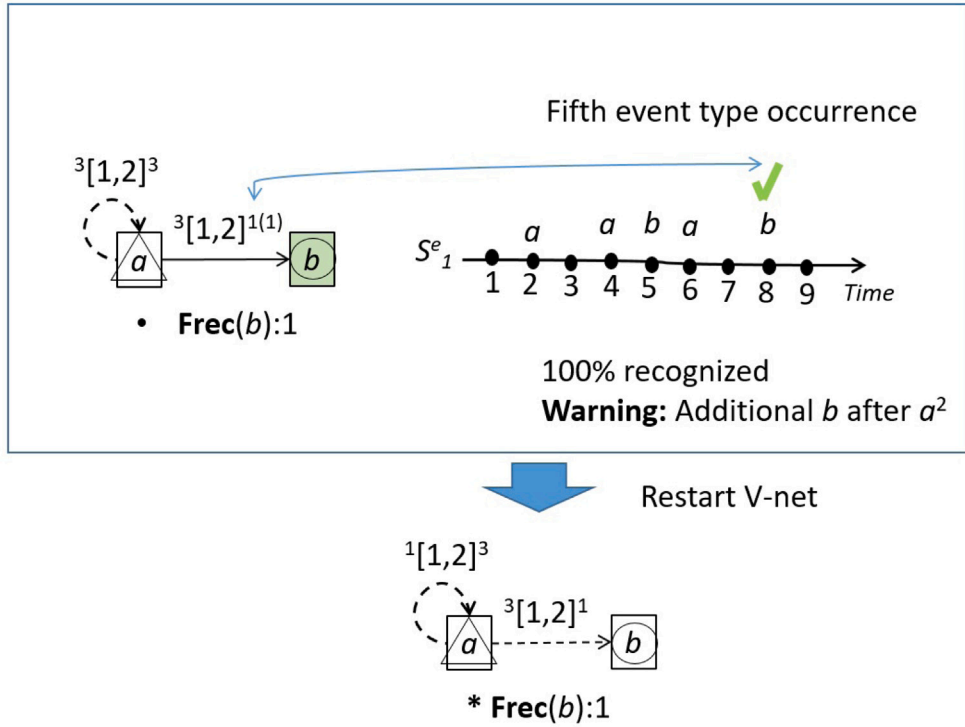
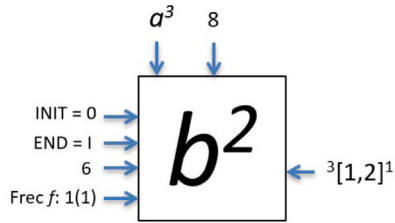


Fig. 21. Fifth timed observation (b,8).

Fig. 22. Representation of the event b^2 as an object.

- $E = \{A, b, N_m, c, P_m, T_m, B\}$
- $\mathcal{T} = \{t_1(A^1 \Rightarrow b^1) : [0, 2]; t_2(A^1 \Rightarrow N_m^1) : [10, 17]; t_3(b^1 \Rightarrow N_m^1) : [10, 15]; t_4(N_m^1 \Rightarrow c^1) : [0, 1]; t_5(c^1 \Rightarrow P_m^1) : [20, 30]; t_6(c^1 \Rightarrow T_m^1) : [20, 30]; t_7(P_m^1 \Rightarrow T_m^1) : [0, 1]; t_8(T_m^1 \Rightarrow P_m^1) : [0, 1]; t_9((P_m^1 \Rightarrow T_m^1) \Rightarrow B^1) : [0, 0]; t_{10}((T_m^1 \Rightarrow P_m^1) \Rightarrow B^1) : [0, 0]\}$

- $\mathbf{Frec} = \{f^{max}(c) : 1; f^{max}(B) : 1\}$
- $tl_{eval} : 58$
- $R = \mathbf{Frec}(B) : 1, A = T_m \wedge P_m \wedge N_m$

4.2.3. V-net for State 3

In State 3: The ascendance of the power, which orchestrates the transition to higher power levels, involves a meticulous sequence of operational actions and concurrent changes in the process variables. First, immediately or at least 1 min after stage 2 ends, the Reactor Power Increase (RPI) is initiated (discrete event RPI_i), marking a deliberate adjustment of the reactor parameters to gradually increase the power output until the full power is arranged (discrete event RPI_f). Operators fine-tune control rods and other critical parameters to modulate nuclear fission rates, with a time restriction of 5 to 10 min between successive RPI events (summarizing from RPI_i to RPI_f). Subsequently, the Steam Generation Initiation (SGI) starts (event SGI),

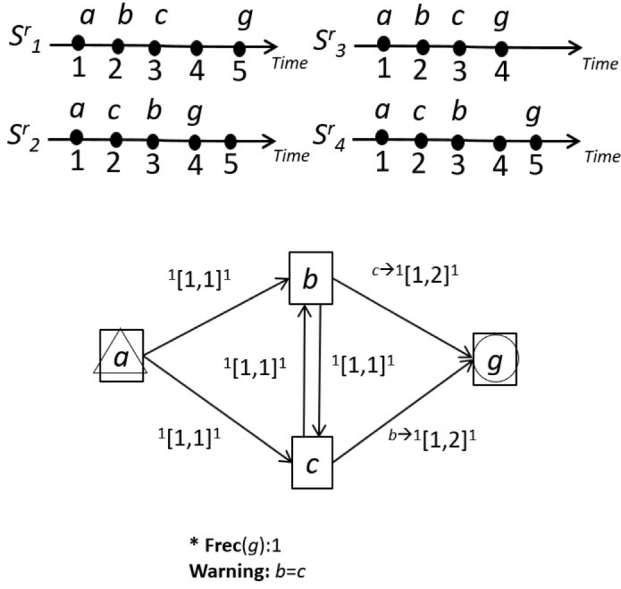
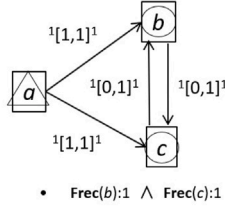


Fig. 23. Example 3. V-net with 4 events.



POSSIBLE OCCURRENCE MATRIX

Matrix 1				Matrix 2				Matrix 3			
	a	b	c		a	b	c		a	b	c
INIT	1	0	0	INIT	1	0	0	INIT	1	0	0
END	0	1	1	Oc1	0	1	0	Oc1	0	0	1
				END	0	0	1	END	0	1	0

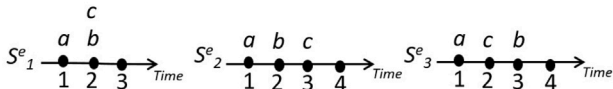


Fig. 24. Example 4 of V-net with its occurrence matrices.

harnessing thermal energy through heat transfer processes, commencing within 10 to 15 min after RPI_f . Following this, Turbine Activation (TA) comes into play (event TA), converting steam's thermal energy into mechanical power to generate electricity, typically within 15 to 20 min post-SGI. Concurrently to TA or within a maximum of 1 min afterward, significant changes unfold in process variables: Temperature (T) surges above 500°C as reactor power and steam generation intensify (from T_m to T_h), Pressure (P) climbs beyond 150 bar with mounting steam pressure (from P_m to P_h). The last process variable to transition to high levels (T_h, P_h, N_h) is typically the neutron flux (N), this Neutron Flux (N) transition from moderate to high level exceeds 5×10^{15} neutrons/cm²/s. Therefore, the time restriction after the temperature (T_h) or pressure (P_h) variables reach high levels to arrange the neutron flux (N_h) typically ranges from 5 to 10 min. These evolutions, unfolding within designated time-frames, guarantee a meticulously controlled and steady progression toward elevated power

levels, thereby enhancing both efficiency and safety in reactor operations. Fig. 31 presents this state's representative sequences of events. Furthermore, Fig. 32 presents the graphical representation of the V-net to supervise this State 3. Note that now, event C is the final event of the V-net and the transition event to the next state.

V_{N3} is defined by

- $E = \{B, RPI_i, RPI_f, SGI, TA, P_h, T_h, N_h, C\}$
- $\mathcal{T} = \{t_1(B^1 \Rightarrow RPI_i^1) : [0, 1]; t_2(B^1 \Rightarrow RPI_f^1) : [5, 11]; t_3(RPI_i^1 \Rightarrow RPI_f^1) : [5, 10]; t_4(RPI_f^1 \Rightarrow SGI^1) : [10, 15]; t_5(SGI^1 \Rightarrow TA^1) : [15, 20]; t_6(TA^1 \Rightarrow P_h^1) : [0, 2]; t_7(TA^1 \Rightarrow T_h^1) : [0, 2]; t_8(P_h^1 \Rightarrow T_h^1) : [0, 2]; t_9(T_h^1 \Rightarrow P_h^1) : [0, 2]; t_{10}((P_h^1 \rightarrow T_h^1) \Rightarrow N_h^1) : [5, 10]; t_{11}((T_h^1 \rightarrow P_h^1) \Rightarrow N_h^1) : [5, 10]; t_{12}(N_h^1 \Rightarrow C^1) : [0, 0]\}$
- $\text{Frec} = \{f^{max}(SGI) : 1; f^{max}(C) : 1\}$
- $tl_{eval} : 59$
- $R = \text{Frec}(C) : 1, C = T_h \wedge P_h \wedge N_h$

4.2.4. V-net for State 4

In State 4: Full Power Operation, the event sequence encompasses crucial operational actions and corresponding transitions in process variables, ensuring sustained power generation while upholding operational safety. The time restriction between the finalization of State 3 and the initiation of Grid Connection typically ranges from 30 to 45 min. Firstly, the connection to the grid (event GC) is initiated, marking the synchronization of the nuclear power plant with the electrical grid for full power generation. This pivotal event typically occurs as the initial step in the sequence, signifying the plant's readiness to supply electricity to the grid. The time restriction between Grid Connection and subsequent events depends on plant-specific protocols and grid synchronization requirements, ensuring a seamless transition to full power operation. Following Grid Connection, Full Power Generation (event FPG) ensues, as the reactor ramps up to its maximum power output capacity. This event represents the culmination of the reactor startup process, with the plant operating at optimal power levels to meet electricity demand. The time restriction between Grid Connection and Full Power Generation typically ranges from 5 to 10 min, allowing for reactor stabilization and power escalation. Continuous Monitoring (event CM) is then initiated, enabling plant operators to oversee and manage reactor operations in real time. This event ensures ongoing monitoring of process variables and operational parameters to maintain system integrity and respond promptly to any deviations or anomalies. The time restriction between Full Power Generation and Continuous Monitoring typically ranges from 1 to 5 min, ensuring immediate monitoring and assessment of reactor performance. During State 4, the transition from high levels (T_h, P_h, N_h) to medium levels (T_m^2, P_m^2, N_m^2) in temperature, pressure, and neutron flux occurs gradually as the reactor stabilizes at full power operation. This transition reflects the equilibrium reached in reactor operations, with process variables stabilizing within operational parameters. The time restriction for this transition typically occurs concurrently with the initiation of Continuous Monitoring, ensuring synchronized monitoring and adjustment of process variables during full power operation. After temperature (T) and pressure (P) have reached medium levels, neutron flux (N) follows suit, typically stabilizing at medium level within 5 to 10 min, although specific timeframes may vary depending on reactor dynamics and control strategies. Fig. 33 presents the representative sequences of events of this state. Furthermore, Fig. 34 presents the graphical representation of the V-net to supervise this State 4. Note that D is the final event of the V-net and the transition event to the final state (5), and the events T_m^2, P_m^2, N_m^2 express that these have occurred for the second time at reactor startup.

V_{N4} is defined by

- $E = \{C, GC, FPG, CM, P_m, T_m, N_m, D\}$

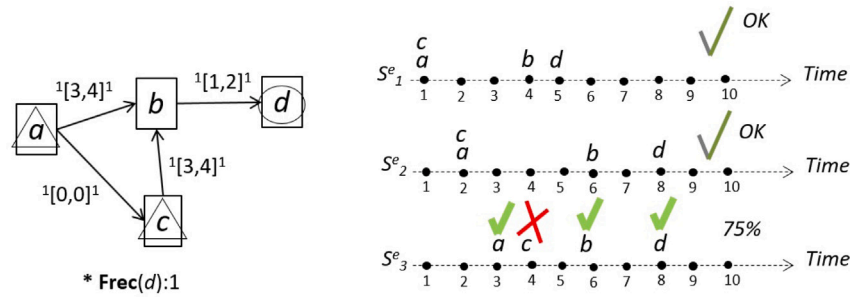


Fig. 25. Example 5 of V-net and its partial recognition of a V-net.

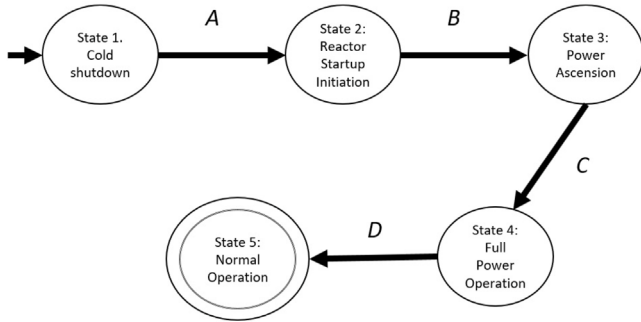


Fig. 26. Automata for a Reactor Startup.

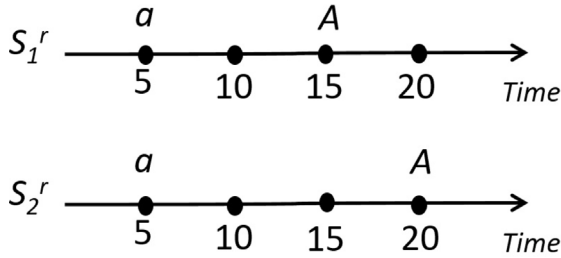
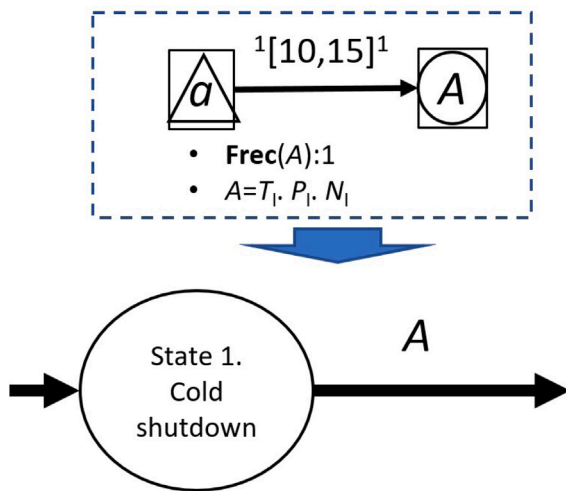


Fig. 27. Representative event sequences for State 1.

Fig. 28. V-net V_{N1} that supervises State 1.

• $\mathcal{T} = \{t_1(C^1 \Rightarrow GC^1) : [30, 45]; t_2(GC^1 \Rightarrow FPG^1) : [5, 10]; t_3(FPG^1 \Rightarrow CM^1) : [1, 5]; t_4(CM^1 \Rightarrow P_m^2) : [0, 0]; t_5(CM^1 \Rightarrow T_m^2) : [0, 0]; t_6(P_m^2 \Rightarrow T_m^2) : [0, 2]; t_7(T_m^2 \Rightarrow P_m^2) : [0, 2]; t_8((P_m^2 \Rightarrow$

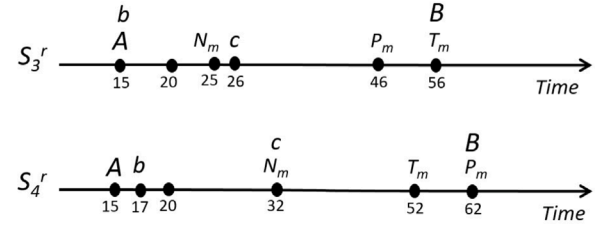


Fig. 29. Representative event sequences for State 2.

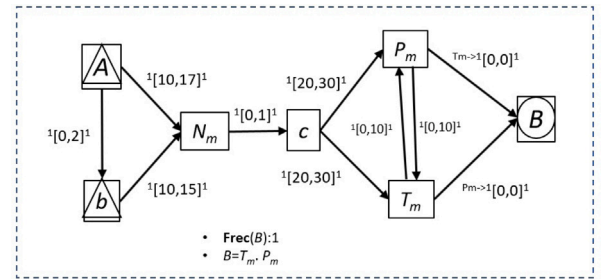
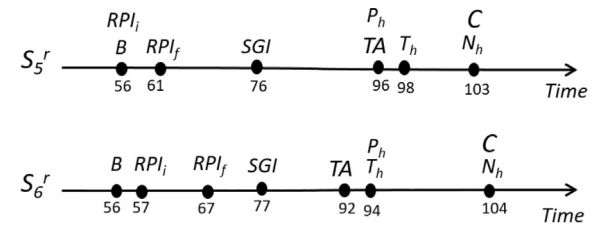
Fig. 30. V-net V_{N2} that supervises State 2.

Fig. 31. Representative event sequences for State 3.

$T_m^2) \Rightarrow N_m^2) : [5, 10]; t_9((T_m^2 \Rightarrow P_m^2) \Rightarrow N_m^2) : [5, 10]; t_{10}(N_m^2 \Rightarrow D^1) : [0, 0]\}$

• $\mathbf{Frec} = \{f^{max}(P_m) : 2; f^{max}(T_m) : 2; f^{max}(N_m) : 2; f^{max}(D) : 1\}$
 • $tl_{eval} : 72$
 • $R = \mathbf{Frec}(D) : 1, D = T_m^2 \wedge P_m^2 \wedge N_m^2$

4.2.5. V-net for State 5

In this case study, State 5 could not necessitate a V-net for the final of this startup, as it signifies a stable operational phase where

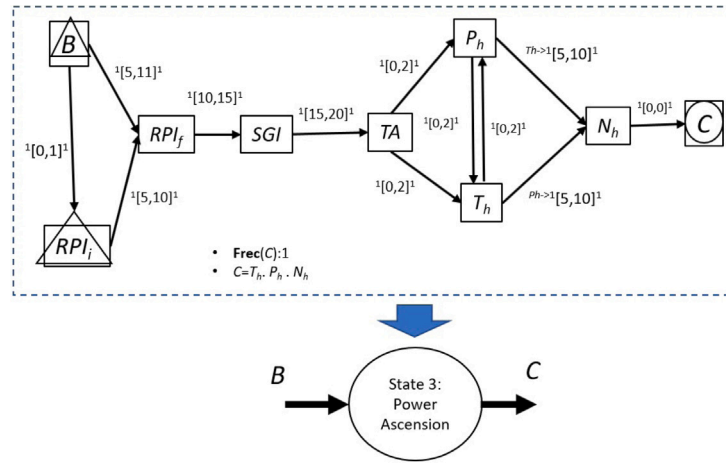
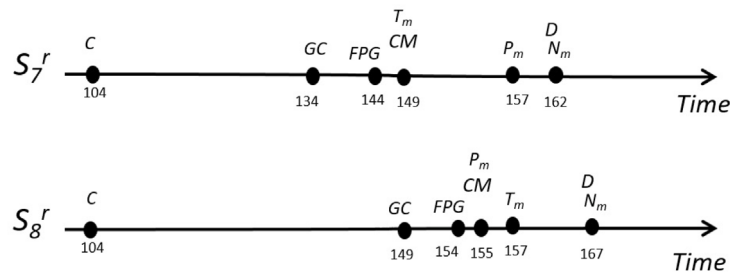
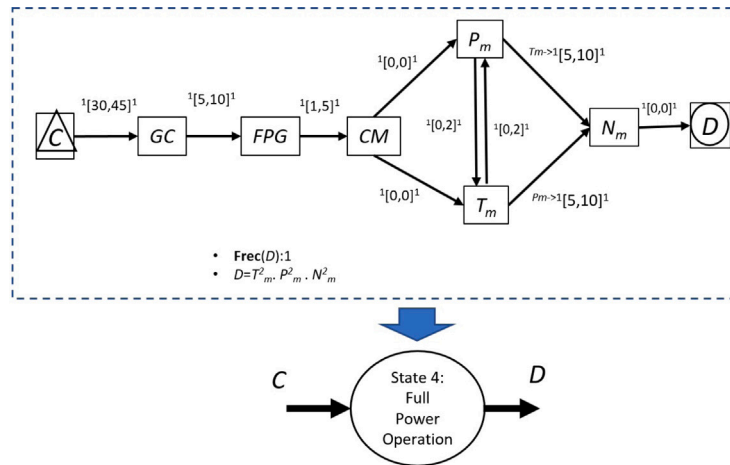
Fig. 32. V-net V_{N3} that supervises State 3.

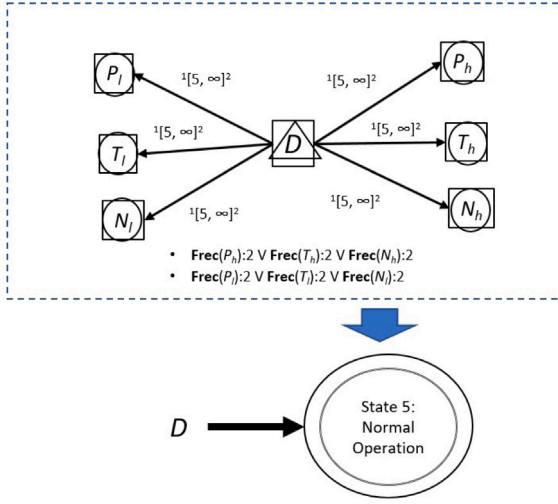
Fig. 33. Representative event sequences for State 4.

Fig. 34. V-net V_{N4} that supervises State 4.

the system has reached a state of equilibrium and routine functionality. The reactor system operates within established parameters in this state, with no significant changes or transitions occurring. Therefore, there is no need for continuous monitoring or dynamic analysis, which is typically facilitated by V-nets, as the system remains stable and predictable. Instead, State 5 primarily involves routine maintenance, periodic inspections, and preparedness for emergency responses to ensure ongoing operational integrity and safety. However, V-nets could be used to analyze the sequence of alarms and determine whether a hazardous situation is active. An additional V-net will be developed to monitor possible alarms generated by high levels of process variables ($T_h, P_h, N_h, T_l, P_l, N_l$). Fig. 35 presents the graphical representation of the V-net to supervise this state 5. Note that after the event D

appears a divergence of six possible events that could occur ($T_h^2, P_h^2, N_h^2, T_l^2, P_l^2, N_l^2$). This indicates that an increase or decrease in the process variables is assumed as a possible failure or abnormal situation to be managed. Now, the time restriction $^1[5, \infty]^2$ indicates that after the reactor is in a stable state, 5 (five) minutes after this process could present some variation in any of the variables. The symbol ∞ (infinity) indicates that there is no time restriction for these events to occur. V_{N5} is defined by

- $E = \{D, P_h, T_h, N_h, P_l, T_l, N_l\}$
- $\mathcal{T} = \{t_1(D^1 \Rightarrow P_h^2) : [5, \infty]; t_2(D^1 \Rightarrow T_h^2) : [5, \infty]; t_3(D^1 \Rightarrow N_h^2) : [5, \infty]; t_4(D^1 \Rightarrow P_l^2) : [5, \infty]; t_5(D^1 \Rightarrow T_l^2) : [5, \infty]; t_6(D^1 \Rightarrow N_l^2) : [5, \infty]\}$

Fig. 35. V-net V_{N5} that supervises State 5.

- $\text{Frec} = \{f^{max}(P_h) : 2 \vee f^{max}(T_h) : 2 \vee f^{max}(N_h) : 2 \vee f^{max}(P_l) : 2 \vee f^{max}(T_l) : 2 \vee f^{max}(N_l) : 2\}$
- $t_{eval}^i : \infty$
- $R = T_h^2 \vee P_h^2 \vee N_h^2 \vee T_l^2 \vee P_l^2 \vee N_l^2$

The expression $R = T_h^2 \vee P_h^2 \vee N_h^2 \vee T_l^2 \vee P_l^2 \vee N_l^2$ indicates that if any of the process variables goes to high or low values, the V-net is recognized, giving an alert to the operators. In the start-up process, it is normal that the high or low levels appear in the variables, but after the startup procedure, these levels are abnormal.

5. Result analysis: Event sequence generation and matching with V-net time restrictions across all states

In this section, we provide a detailed analysis of the event sequences generated for the five V-nets corresponding to the reactor startup process. The goal of the analysis is to generate more than 100,000 event sequences using predefined event types and time constraints for each V-net. The sequences are then analyzed to ensure that they conform to the time restrictions of each V-net. Finally, a Monte Carlo method is applied to select a few representative event sequences for further evaluation.

5.1. Overview of algorithms

The event sequence generation and matching process is divided into the following key steps:

1. ****Generate more than 100,000 event sequences**** using the event types defined in each of the five V-nets and the time constraints associated with each transition.
2. ****Validate each event sequence**** against the time constraints of the V-nets to ensure that the generated sequences adhere to the operational requirements.
3. ****Apply a Monte Carlo sampling method**** to select representative event sequences for further evaluation.

The algorithms implemented to perform these tasks are described below.

5.2. Algorithm for event sequence generation

The first algorithm generates more than 100,000 event sequences, ensuring that all event types for each of the five V-nets are included in the sequences. The algorithm is described step by step below.

Algorithm 1: Event Sequence Generation Algorithm for All V-nets

Data: Event types and time restrictions for all five V-nets

Result: Generate more than 100,000 valid event sequences containing all event types.

Function GenerateEventSequence():

Initialize an empty sequence $S \leftarrow []$;

Set current_time $\leftarrow 0$;

foreach V-net V_n in [V-net 1, V-net 2, V-net 3, V-net 4, V-net 5]

do

foreach event e_i in V-net **do**

Retrieve time restriction for event e_i :

$T_{range} \leftarrow \text{time_restriction}(e_i)$;

Generate random event time: $t_{event} \leftarrow$

$\text{random}(\text{current_time} + T_{range}[0], \text{current_time} + T_{range}[1])$

;

Append (e_i, t_{event}) to sequence S ;

Update current_time $\leftarrow t_{event}$;

return S

Generate more than 100,000 event sequences:

$\text{event_sequences} \leftarrow [\text{GenerateEventSequence}() \text{ for } i \in \text{range}(100001)]$

Explanation:

The function GenerateEventSequence() generates a valid event sequence for each of the five V-nets by sequentially generating event types. For each event type in each V-net, the time restrictions are retrieved, and the event time is randomly generated within the specified time range. The resulting event sequence contains all event types and their corresponding timestamps. This process is repeated to generate more than 100,000 valid event sequences.

5.3. Algorithm for matching event sequences with V-net time restrictions

The next step is to validate the generated event sequences by checking their conformity with the time restrictions of the V-nets. Each sequence is checked to ensure that all transitions between events adhere to the time intervals defined in the V-net.

Algorithm 2: Matching Event Sequences with V-net Time Restrictions

Data: Generated event sequences, V-net time restrictions

Result: Filter valid event sequences matching V-net time constraints.

Function MatchWithVnet(sequence):

foreach $(e_i, t_i), (e_{i+1}, t_{i+1})$ in sequence **do**

Retrieve time constraint:

$T_{expected} \leftarrow \text{getTimeConstraint}(e_i, e_{i+1})$;

if $t_{i+1} - t_i$ is not within $T_{expected}$ **then**

return False;

return True

Filter valid sequences that match V-net time restrictions:

$\text{valid_sequences} \leftarrow [S \text{ for } S \in$

$\text{event_sequences} \text{ if } \text{MatchWithVnet}(S)]$

Explanation:

The function MatchWithVnet() takes each generated sequence and checks the time constraints between consecutive events. If any event transition does not satisfy the time constraints specified in the V-net, the sequence is discarded. Only sequences that adhere to the V-net constraints are considered valid. These valid sequences are stored in the list valid_sequences for further analysis.

5.4. Monte Carlo sampling algorithm

Once the valid event sequences are filtered, a Monte Carlo sampling method is applied to select a few representative sequences for further

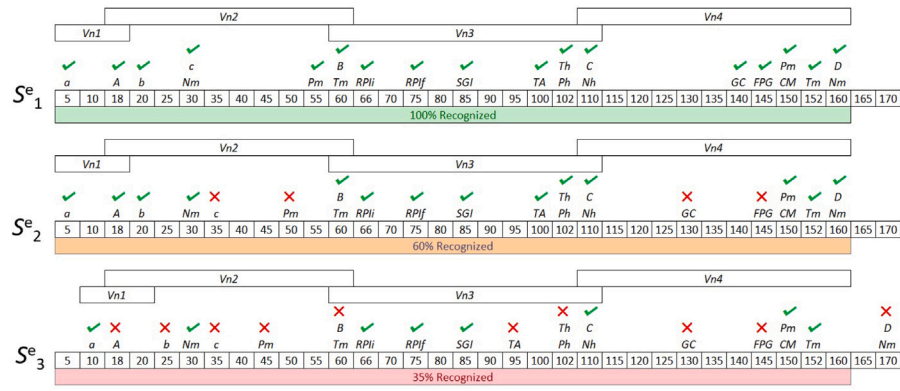


Fig. 36. Evaluation of the V-nets with S_1^e , S_2^e , and S_3^e across all states.

evaluation. This method helps in randomly selecting sequences that can represent the overall set of valid sequences.

Algorithm 3: Monte Carlo Sampling for Event Sequences

Data: Valid event sequences

Result: Select a few representative event sequences for further evaluation.

Function MonteCarloSelection(*sequences*, *num_samples*):

 return random.sample(*sequences*, *num_samples*)

Select 3 representative sequences for evaluation:

selected_sequences ← MonteCarloSelection(*valid_sequences*, 3)

Explanation:

The function MonteCarloSelection() selects a specified number of event sequences from the valid sequences using random sampling. In this case, three representative sequences are chosen for further evaluation. These sequences are selected randomly to ensure diversity and comprehensive coverage of the reactor startup process.

5.5. Results and discussion

The algorithms described above successfully generated more than 100,000 event sequences based on the predefined event types and time restrictions for the five V-nets. After applying the matching algorithm, only sequences that adhered to the V-net time constraints were retained. A Monte Carlo sampling method was then employed to select three representative sequences for further evaluation. Fig. 36 visualizes the evaluation of the selected sequences with the V-nets across all five states. The figure illustrates the time constraints imposed by the V-nets, the occurrence of events in the generated sequences, and the percentage of matches between the sequences and the V-net time restrictions.

The results for the three selected event sequences are as follows:

- Sequence S_1^e achieved a 100% match with the V-net time constraints. For instance, the time between event *a* (Initialization of Reactor Systems) and event *A* (Pre-Startup Checks) in V-net State 1 satisfied the constraint $a^1[10, 15]^1A$. Similarly, the transition between *SGI* and *TA* in V-net State 3 occurred within the allowed time interval.
- Sequence S_2^e matched 60% of the V-net constraints. Some temporal deviations occurred in V-net State 2, where event *c* (Coolant System Activation) occurred earlier than allowed, leading to subsequent deviations in pressure (P_m) and temperature (T_m).
- Sequence S_3^e matched only 35% of the V-net constraints. Notably, event *b* (Control Rod Withdrawal) in V-net State 2 occurred much later than allowed, resulting in a significant mismatch with the prescribed time intervals.

These results highlight the importance of time constraints in ensuring the correct sequence of events in the reactor startup process. The V-nets effectively monitor the event sequences, ensuring that all transitions adhere to the predefined operational requirements.

5.6. Opportunities for future development in V-net theory

While V-nets offer substantial advancements in handling fault diagnosis in Discrete Event Systems (DES), the current implementation highlights several exciting opportunities for future improvements. As a new formalism, V-nets present a solid foundation, but their full potential will be realized as the theory evolves and matures. This section discusses the current limitations of V-nets not as drawbacks, but as opportunities for further development, such as the automation of V-net generation, real-world applications beyond simulations, and scalability testing. These opportunities promise to refine the V-net model and solidify its place in industrial fault diagnosis.

5.6.1. Automatic V-net generation algorithms

One of the current challenges is the manual construction of V-nets, which requires detailed system knowledge to define event types, time constraints, and logical predicates. However, this presents a significant opportunity: the development of algorithms that can automatically generate V-nets from system specifications, operational data, or pre-defined event sequences. Automating the V-net creation process would streamline its implementation in complex industrial systems and reduce the need for specialized expertise in constructing V-net models. Future research will focus on creating an algorithm capable of automatically constructing V-nets from representative event sequences and operational data. This will not only simplify the model construction phase but also enable broader adoption of V-nets in industries where quick deployment of fault diagnosis systems is critical. By leveraging machine learning and data-driven approaches, V-net models could be dynamically updated as the system evolves, ensuring that the fault diagnosis framework remains relevant and accurate over time.

5.6.2. Application in real-world industrial processes

Thus far, the application of V-nets has primarily been demonstrated through simulations, such as in the case of the reactor startup process. While simulations are a powerful tool for testing and refining the model, real-world implementation is the next logical step to validate the scalability and effectiveness of V-nets in actual industrial environments. The opportunity here lies in applying V-nets to real processes, where they can demonstrate their capabilities in monitoring and diagnosing complex systems in real time. Implementing V-nets in live industrial processes would provide valuable insights into their performance under real-world conditions, such as varying operational loads, unpredictable disturbances, and unforeseen faults. These implementations will not only prove the robustness and reliability of V-nets but also help identify areas for further refinement. By working closely with industry partners, future studies can test V-nets on a wide range of industrial systems, from manufacturing to energy generation, thereby demonstrating the versatility and scalability of the model.

5.6.3. Scalability testing and optimization

As V-nets are a relatively new formalism, their scalability in large-scale industrial systems has yet to be fully tested. The ability to handle a large number of simultaneous events, time constraints, and complex interactions across multiple subsystems is crucial for their successful deployment in large-scale operations. This limitation offers an exciting opportunity for future research: optimizing V-nets for scalability. Scalability testing will involve applying V-nets to increasingly complex systems, ensuring that their performance remains robust and efficient. Optimizations, such as parallel processing techniques, modular V-net designs, and the use of distributed computing architectures, can be explored to enhance the scalability of V-nets. This will ensure that V-nets can be deployed in systems with thousands of interacting components, maintaining their ability to handle simultaneous events and time-sensitive operations without incurring excessive computational costs. Additionally, by introducing macro-level V-nets (or “macro nets”), future research can model subsystems as individual V-nets within a larger, overarching system. This modular approach will allow V-nets to manage complex systems more efficiently by breaking them down into smaller, more manageable parts while retaining the overall system’s integrity. Macro nets will not only improve scalability but also facilitate easier maintenance and updates to the model as the system evolves.

5.6.4. Adaptation for stochastic systems

Currently, V-nets excel in deterministic systems where event timings and sequences follow predictable patterns. However, many real-world industrial systems exhibit stochastic behavior, where randomness and uncertainty play a significant role in system dynamics. Adapting V-nets to handle stochastic systems presents an opportunity to expand their applicability to a wider range of industrial processes. Future research can explore hybrid V-net models that integrate stochastic elements, allowing them to account for random faults, varying process times, and other uncertainties inherent in stochastic systems. This adaptation will enable V-nets to remain relevant in environments where randomness is prevalent, such as supply chain management, dynamic scheduling systems, and fault diagnosis in systems with unpredictable workloads or failures. Hybrid V-nets could incorporate probabilistic transitions and dynamic time constraints, making them more versatile in handling both deterministic and stochastic systems.

5.6.5. Continuous improvement through data integration

Another opportunity for the future development of V-nets lies in their continuous improvement through data integration. As industrial systems become more data-driven, V-nets can leverage real-time data to refine and adapt their models. Integrating V-nets with data analytics platforms, IoT devices, and machine learning algorithms can enable continuous monitoring and adjustment of event sequences and time constraints. By utilizing operational data, V-nets can be continuously updated to reflect the current state of the system, learning from new event sequences and adapting to changing operational conditions. This integration will improve the accuracy and relevance of V-nets in fault diagnosis, making them a dynamic tool that evolves alongside the systems they monitor. Furthermore, data-driven V-nets will reduce the need for manual updates and recalibration, offering a more sustainable and adaptive approach to fault detection. The limitations identified in the current implementation of V-nets are not obstacles but opportunities for future growth and development. The theory behind V-nets holds immense potential for transforming fault diagnosis in industrial systems, but it is still in its early stages. Future research will focus on automating V-net generation, testing their scalability in real-world environments, and adapting them for stochastic systems. These developments will solidify V-nets as a versatile, scalable, and highly effective tool for fault diagnosis in a wide range of industrial applications. By embracing these opportunities, the V-net framework will continue to evolve, offering more robust and efficient solutions to the challenges faced by modern industrial systems.

6. Conclusion

This paper introduced and developed the foundational theory of V-nets, a novel formalism designed to address critical challenges in fault diagnosis and event sequence management within Discrete Event Systems (DES). V-nets represent a significant advancement over traditional models such as Petri nets and automata, particularly in their ability to manage simultaneous occurrences, minimize false positives, and improve the recognition of partial event sequences. Through the analysis of the reactor startup sequence in a Pressurized Water Reactor (PWR), V-nets have demonstrated their potential as a robust tool for monitoring and diagnosing complex industrial systems. The results highlighted how V-nets enable more accurate supervision of event sequences and time constraints across different stages of the startup process. With the generation of over 100,000 event sequences and the application of Monte Carlo methods for sequence evaluation, this study showcased the scalability of V-nets in handling extensive event data while ensuring system reliability.

Key findings from this study include:

- V-nets offer a flexible framework for modeling complex temporal constraints and simultaneous events, enabling precise fault diagnosis in systems characterized by dynamic and interdependent event sequences.
- The Monte Carlo-based evaluation method demonstrated the effectiveness of V-nets in selecting representative event sequences, allowing for more efficient system monitoring and fault detection.
- By applying V-nets to the reactor startup sequence, this study highlighted their adaptability in handling both deterministic and complex event processes, ensuring compliance with time restrictions and operational conditions.

The significance of this work lies in its potential to transform how event sequences are modeled and diagnosed in industrial systems. V-nets provide a much-needed improvement over existing methods by reducing false positives, accommodating simultaneous events, and enabling partial recognition of sequences. The scalability of the V-net model, demonstrated through the generation of thousands of event sequences, suggests that it can be effectively applied to a wide range of industrial applications beyond the case study presented.

6.1. Opportunities for future work

While the current study has laid a strong theoretical foundation for V-nets, several exciting opportunities for future work remain. Key areas for future research include:

- **Automation of V-net Construction:** Developing algorithms for the automatic generation of V-nets based on event sequence data and system specifications will streamline the implementation process and make V-nets accessible to a broader range of applications.
- **Real-world Implementation:** While the current work focuses on simulations, future research will involve applying V-nets in real-world industrial systems. This will provide insights into their scalability, robustness, and adaptability under varying operational conditions.
- **Scalability and Optimization:** Testing V-nets in large-scale systems will allow for the refinement of the model to ensure it can efficiently manage the complexity of large industrial systems, potentially through parallel processing and modular designs (macro nets).
- **Handling Stochastic Systems:** Future work will focus on adapting V-nets to handle stochastic systems where uncertainty and randomness are inherent in system dynamics.

6.2. Future directions and introduction to part II

In Part II of this series, we will explore the practical applications of V-nets through in-depth case studies and examples that highlight their effectiveness in modeling dynamic processes beyond traditional state machines. This upcoming paper will delve deeper into various properties of V-nets, such as handling simultaneous occurrences, convergence, and divergence, and their influence on event sequences. Through tangible examples and real-world scenarios, we will illustrate how V-nets can be used to capture complex temporal constraints, event repetitions, and event convergence and divergence. The goal of Part II is to provide researchers, engineers, and practitioners with a deeper understanding of how V-nets can be practically applied in diagnosing and managing complex systems. By bridging theoretical concepts with detailed examples, we aim to offer a comprehensive guide on implementing V-nets in real-world environments, ensuring their broader adoption in industries requiring advanced fault diagnosis and event sequence management. The evolution of V-nets as a powerful tool for fault diagnosis and dynamic process modeling will undoubtedly open new avenues for research and innovation in the field of DES, ensuring that industrial systems operate with greater efficiency, reliability, and safety.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No data was used for the research described in the article.

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